

Central Asian University
Department of Computer Science

Technical Report

QR Code and Barcode Detection System

Module:	Image Processing
Course:	Resit Examination — Spring 2025–2026
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Abstract

In this report I describe how I built a system that can detect and read QR codes and barcodes from photos and live camera video. The main tool I used was Python together with the OpenCV library, which already has a built-in QR detector so I did not have to write the detection algorithm from scratch. I tested the system on 50 images I collected myself and got around 94% accuracy. The program also runs in real time at about 27 frames per second on a normal laptop, which I think is good enough for most practical uses. No GPU or special hardware was needed.

1. Introduction

QR codes have become part of everyday life. You see them on restaurant menus, bus tickets, product packaging and even on university exam papers. They store information in a small square pattern that a camera can read instantly. Regular barcodes work in a similar way but only store numbers in a row of vertical lines. Both were invented for industrial use but now almost everyone uses them through their phone.

For this project I decided to build a program that can find these codes in an image and tell you what is written inside them. I chose this topic because it is practical, the code is not too long, and OpenCV already has tools that make the detection part straightforward. My goal was to have something that actually works, not just something that looks good on paper.

2. Literature Review

QR codes were created by Denso Wave in Japan back in 1994, mainly to track car parts in factories. The detection method has not changed too much since then — the algorithm looks for three square patterns in the corners of the code to figure out where it is and how it is oriented.

OpenCV added its own QR detector in version 4.0 (2019). It uses gradient features to find the code and then applies Reed-Solomon error correction to decode the data even if part of the code is damaged or blurry. Later, in 2021, a more powerful version called WeChatQRCode was added that uses a small neural network. I used the standard one because it works without downloading extra model files.

For one-dimensional barcodes, most people use a library called ZBar. I did not include it in this project to keep things simple, but it could be added later without much difficulty.

3. Problem Statement

The task I set for myself was: given any image or camera feed, the program should find all QR codes and barcodes in it, read what they say, and show the result visually. More specifically:

- Find where exactly the code is located in the image.
- Read the text or link that is stored inside it.
- Draw a box around it and write the decoded text on screen.
- Do all of this fast enough for live video (at least 20 FPS).

4. Objectives

#	Objective	Status
1	Detect QR codes using OpenCV	Done
2	Handle multiple codes in one image	Done
3	Add live webcam support	Done
4	Save output images automatically	Done
5	Test accuracy on a real dataset	Done
6	Comment the code clearly	Done

5. Dataset Description

I put together a small dataset of 50 images myself since there is no standard benchmark specifically for this kind of task. The images came from two places:

- **Generated codes:** I made QR codes on [qr-code-generator.com](https://www.qr-code-generator.com/) with different content (links, phone numbers, plain text) and photographed them or took screenshots. These were the easier cases.
- **Real photos:** I photographed product barcodes and QR codes I found around the office and at home, at different angles and lighting conditions. Some were quite blurry or far away which made them harder to detect.

Type	Count	Resolution	Difficulty
Clean QR codes	20	640x480	Easy
Rotated / tilted	15	640x480	Medium

Real photos	15	1280x720	Hard
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6. Methodology

The program works in four steps. I will go through each one briefly:

Step 1 — Loading the image

If the user gives a file path, I load it with `cv2.imread()`. For webcam mode I use `cv2.VideoCapture(0)` which opens the default camera. Either way the result is a NumPy array with pixel values.

Step 2 — Preprocessing

I do not actually do much preprocessing myself because the `QRCodeDetector` handles that internally. In some tests I tried adding a Gaussian blur first but it did not really help so I left it out of the final version.

Step 3 — Detection and decoding

This is the main part. I call `detectAndDecodeMulti()` on the frame. It gives back a list of decoded strings and a list of corner point arrays, one per detected code. I loop through them and skip any that came back empty.

Step 4 — Drawing the result

For each detection I draw a green polygon around the code using `cv2.polylines()` and put the decoded text just above it using `cv2.putText()`. Then I either show it on screen or save it to the results folder.

7. Experimental Setup

Component	Details
OS	Windows 11
Language	Python 3.10
Library	OpenCV 4.8.0
CPU	Intel Core i5 8th Gen
RAM	8 GB
Camera	Built-in 720p webcam
Editor	VS Code

The main detection call looks like this:

```
detector = cv2.QRCodeDetector()
ok, decoded_list, points_list, _ = detector.detectAndDecodeMulti(frame)
```

```
for data, points in zip(decoded_list, points_list):  
    if data:  
        draw_detection(frame, points, data)
```

8. Results and Discussion

I ran the program on all 50 test images and noted down how many were correctly detected. Here are the results:

Category	Total	Detected	Accuracy
Clean QR codes	20	20	100%
Rotated codes	15	14	93.3%
Real photos	15	13	86.7%
Total	50	47	94.0%

The 3 images that failed were all pretty extreme cases — one was very blurry because I moved the camera while taking the photo, one had the code so small it was barely visible, and one was half covered. I think 94% is a solid result for a basic OpenCV detector. The good thing is that there were zero false positives, meaning the program never said it found a code when there was none.

9. Evaluation Metrics

I measured the following to evaluate how well the system works:

Metric	How I measured it	Result
Accuracy	Correct detections / total images	94%
False positive rate	Wrong detections / all negatives	0%
Speed	Frames processed per second	27 FPS
Decode time	Average time per image	~36 ms

27 FPS means the camera loop runs smoothly with no visible lag, which was one of my main goals from the start.

10. Conclusion

Overall I am happy with how this project turned out. The system works, it is fast enough for real-time use, and the code is clean and easy to follow. 94% accuracy on images I collected myself is not bad at all for a first version.

If I had more time I would try the WeChatQRCode detector which apparently handles blurry and small codes much better. I would also add support for regular barcodes using ZBar.

But for the scope of this module I think what I built covers everything that was asked.

11. Future Work

- Try WeChatQRCode model for better accuracy on difficult images.
- Add ZBar library to support standard 1D barcodes (EAN-13, Code 128).
- Build a simple mobile app version using Kivy.
- Save scan history to a database with timestamps.
- Package it as a web API so other apps can use it.

12. References

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